Ehsan Ghoreishi

Virginia Tech, Blacksburg, VA, U.S.

LinkedIn   Website   Google Scholar

EDUCATION

Virginia Tech — **Blacksburg, VA, U.S.**
Ph.D. in Computer Engineering — 2024-present

Virginia Tech — **Blacksburg, VA, U.S.**
M.Sc. in Computer Engineering — 2024

Isfahan University of Technology — **Isfahan, Iran**
B.Sc. in Electrical Engineering — 2021

RESEARCH INTERESTS

Wireless Networks, Machine Learning, Reinforcement Learning, Real-Time Systems, Optimization

RESEARCH EXPERIENCE

Design an intelligent NextG scheduler for URLLC/eMBB multiplexing — 2022–present
Complex Network & Security Research Lab, Virginia Tech

- **Cyrus**: Designed an O-RAN–compliant DRL-based puncturing scheduler for URLLC/eMBB multiplexing.
- **Cyrus+**: Advanced DRL-based puncturing scheduler achieving real-time operation ($\sim$125 μs) to maximize eMBB goodput while ensuring URLLC latency guarantees.
- **PuRe**: Designed a learning-based framework that (a) predicts URLLC packet arrival patterns using an LSTM model, and (b) applies DRL-based puncturing on eMBB transmissions to maximize goodput while ensuring URLLC latency guarantees.
- Conducted a survey on how machine learning techniques can enhance eMBB/URLLC multiplexing.

Satellite Communication — 2025–present
Complex Network & Security Research Lab, Virginia Tech

- Investigated LSTM-driven CQI prediction methods to enable adaptive resource block allocation under long LEO propagation delays.
- Addressed the challenge of short satellite dwell times ($\sim$15 s in Starlink) by exploring lightweight deep learning models for real-time training and inference.

Complex-Valued Neural Network for Wireless Modulation Classification — 2020–2021
Signal Processing Research Center, Isfahan University of Technology

- Designed and implemented a complex-valued CNN model for automatic wireless modulation classification.
- Leveraged in-phase/quadrature (I/Q) signal representations to improve feature extraction and classification accuracy.

Designed and Analyzed Nonlinear Effects in Remote Wire Monitoring — 2020–2021
Optics Research Center, Isfahan University of Technology

- Developed optical sensing techniques to remotely monitor structural wires under nonlinear effects.
- Modeled and analyzed nonlinear optical responses to enhance detection sensitivity and reliability.

PUBLICATIONS

Patent

Ehsan Ghoreishi, Bahman Abolhassani, Y. Thomas Hou, “A Method for Multiplexing Ultra Low Latency Data with Other Data over the Same Radio Frame,” U.S. Patent Application.

Papers

Ehsan Ghoreishi, Bahman Abolhassani, Yan Huang, Shiva Acharya, Wenjing Lou, Y. Thomas Hou, “Cyrus: A DRL-based Puncturing Solution to URLLC/eMBB Multiplexing in O-RAN," *in Proc. International Conference on Computer Communications and Networks (ICCCN)*, 9 pages, Hawaii, USA, 2024. (Invited Paper)

Ehsan Ghoreishi, Bahman Abolhassani, Yan Huang, Shiva Acharya, Wenjing Lou, Y. Thomas Hou, “Cyrus+: A Real-Time DRL-based Puncturing Solution to URLLC/eMBB Multiplexing in O-RAN," *IEEE Transactions on Machine Learning in Communications and Networking*.

Ehsan Ghoreishi, Bahman Abolhassani, Wenjing Lou, Y. Thomas Hou, “PuRe: Learning-based Joint Puncturing and Reservation for Efficient URLLC/eMBB Multiplexing," *in Proc. IEEE International Conference on Computer Communications (INFOCOM)*), 10 pages, Tokyo, Japan, 2026.

Ehsan Ghoreishi, Shiva Acharya, Wenjing Lou, Y. Thomas Hou "Enhancing 5G Efficiency: How Machine Learning is Revolutionizing eMBB and URLLC Puncturing." (To be submitted to *IEEE Wireless Communications*)

Poster...

Ehsan Ghoreishi, Bahman Abolhassani, Yan Huang, Shiva Acharya, Wenjing Lou, Y. Thomas Hou "Cyrus: A DRL-based Puncturing Solution to URLLC/eMBB Multiplexing in O-RAN," *Office of Naval Research (ONR) MURI Meeting*, Washington D.C., USA, 2024.

AWARDS AND ACHIEVEMENTS

CCI SWVA Cyber Innovation Scholar 2025
Recipient, Commonwealth Cyber Initiative – SWVA (**Funding Number:** 556194)

Nationwide University Entrance Exam – Mathematical Physics 2017
Ranked 90^{th} among 200,000 participants nationwide

TEACHING EXPERIENCE

The Graduate Teaching Assistant in Introduction to Computer Networking 2022
Bradley Dept. of Electrical and Computer Engineering, Virginia Tech

The Teaching Assistant in Principles of Electronics 2021
Electrical and Computer Engineering Dept., Isfahan University of Technology

The Teaching Assistant in Electric Circuits II 2021
Electrical and Computer Engineering Dept., Isfahan University of Technology

Course Instructor in MATLAB 2022
Electrical and Computer Engineering Dept., Isfahan University of Technology

SKILLS

Programming Languages: Python, MATLAB, C/C++, Java
Machine Learning & Data Science: PyTorch, TF-Agents, TensorFlow
Engineering Tools: MATLAB, Simulink, Arduino, AutoDesk, AutoCad
Web: Spring, Flask, HTML5, CSS, Bootstrap

PROFESSIONAL MEMBERSHIPS

Institute of Electrical and Electronics Engineers (IEEE) 2022-present

IEEE Communications Society (ComSoc) 2022-present